Soccer Penalty Kick Direction Prediction from Pre-Contact Video Using Deep Learning

Final semester project report | Abdul-Azeem Opakunle and Zackary Butz

Evaluation: verified five-fold comparison; source implementation 47a9824; published results e13c942; demo correction 54dfd26.

Abstract

This project studies whether the direction of a soccer penalty kick can be inferred before ball contact from video, using labels defined from the goalkeeper's perspective. The original goal was left/center/right prediction; however, the available labeled data contained only one center example. Consequently, the reported controlled evaluation is explicitly binary (left versus right), with 119 eligible kicks (66 left, 53 right). We compare a majority-class predictor, a pose-feature logistic-regression baseline, a pretrained MobileNetV3-Small visual encoder with a gated recurrent unit (GRU), and a single-frame MobileNetV3-Small classifier at pre-contact horizons of 1.0, 0.5 and 0.2 seconds. Five grouped cross-validation folds yield pooled out-of-fold predictions for each configuration. The strongest result is pose logistic regression at 0.2 seconds (63.87% accuracy; 63.36% macro-F1). The GRU slightly exceeds the single-frame model at 0.2 seconds (53.78% versus 51.26% accuracy), but not at earlier horizons. Results suggest that additional temporal modeling is not consistently beneficial in this limited-data setting. No statistical-significance or three-class deployment claim is made.

1. Introduction and Motivation

Goalkeepers have very little time to respond to a penalty kick. Visual cues from the kicker's approach, posture and movement may provide information before the ball is struck. Predicting direction early is therefore a concrete computer-vision task with possible relevance to sports analytics, training and decision support. The central question is not merely whether a neural network can fit the training clips, but whether temporal video modeling improves held-out prediction compared with simpler alternatives.

This project implements an end-to-end workflow: labeled pre-contact video preparation, visual and pose-based feature processing, model training, grouped evaluation, metric aggregation and a command-line prediction demonstration. The controlled comparison requested by the instructor is MobileNetV3-Small with GRU temporal aggregation versus MobileNetV3-Small using a single frame.

2. Problem Definition

Input: a pre-contact video clip of a penalty attempt, sampled relative to the estimated ball-contact moment. Output: a categorical predicted kick direction, expressed from the goalkeeper's perspective. The original design targets left, center and right. The final quantitative experiment instead evaluates left/right because a single center sample cannot support a credible three-class split, model assessment or per-class performance estimate.

Prediction horizons are 1.0, 0.5 and 0.2 seconds before contact. The horizon denotes the latest permitted pre-contact information for a prediction, not the time required to execute inference. This is exploratory research rather than a validated real-time goalkeeper assistance system.

3. Dataset and Preprocessing

3.1 Data composition and provenance

The published Mendeley Data version 2 collection contains 133 penalties and 11,140 pose-tracking frames from collegiate women's soccer training sessions and games [4]. Our evaluated subset comprises 119 eligible labeled kicks: 66 left (55.46%) and 53 right (44.54%). The full available collection contains only one center kick. The final comparison uses the locally prepared video/annotation pipeline. The evidence available for this report does not establish that both initially proposed external sources (SoccerNet and the Mendeley collection) were independently incorporated into these 119 clips; dataset origin, licenses, clip counts before filtering and deduplication must be confirmed against the manifest before final submission. The source README reports that the Mendeley v2 archive contained 132 unique kick IDs for 133 videos, with an identifier mismatch and duplicate tracking rows corrected during indexing. After eligibility checks, 120 clips had all three windows, including one center clip; removing the center example left 119 for binary evaluation.

3.2 Preprocessing and leakage safeguards

The video pipeline produces pre-contact inputs for three horizons; the paired neural comparison decodes 16 frames per kick during timed evaluation. The models use a pretrained MobileNetV3-Small visual backbone. Grouped five-fold cross-validation was used to reduce overlap between related examples across training and held-out folds. The quality of this protection depends on the correctness and completeness of the grouping identifiers (for example, player or match IDs). The precise contact-frame annotation, frame resize, normalization, augmentation, and handling of missing or noisy frames should be transcribed from the committed preprocessing code rather than inferred.

The moderate class imbalance motivates reporting macro-F1 alongside accuracy. The majority baseline exposes the misleadingly acceptable accuracy obtainable by always predicting left, despite failing entirely on right kicks.

4. Deep Learning Methodology

4.1 MobileNetV3-Small visual encoder

MobileNetV3-Small is an efficient convolutional architecture intended for resource-constrained vision workloads [1]. In the controlled neural comparison, an ImageNet-pretrained visual encoder is frozen rather than fully fine-tuned. This reduces trainable parameters and computational cost, but may constrain adaptation to soccer-specific visual cues. The output of the encoder serves as the visual feature representation.

4.2 Temporal GRU classifier

The temporal model processes a sequence of frame embeddings and uses a gated recurrent unit (GRU) to aggregate information across time before classification. GRUs employ learned gates to update hidden

state across sequential inputs [2]. This architecture can represent changes in body orientation or approach dynamics that a single still image may miss. The controlled experiment trains the GRU/classification components while retaining the frozen visual encoder.

4.3 Single-frame classifier and conventional baselines

The single-frame model uses the same MobileNetV3-Small encoder family but predicts from one selected frame using a classification head, without GRU sequence aggregation. A majority-class baseline predicts the most frequent class; a pose-feature logistic-regression baseline evaluates whether a simpler representation captures direction-related information. The GRU and single-frame heads have different capacities; therefore the experiment compares two complete configurations, not a perfectly isolated temporal-information ablation.

4.4 Training specification

Item	Verified setting / reporting note
Training epochs	15 per fold for each neural configuration
Cross-validation	5 grouped folds
Prediction horizons	1.0 s, 0.5 s, 0.2 s
Visual encoder	ImageNet-pretrained MobileNetV3-Small, frozen
Batch size	4 in the recorded comparison command
Seed	42 in the recorded comparison command
Compute	Apple Silicon MPS in the recorded comparison command
Loss, optimizer, learning rate, dropout	Class-weighted cross-entropy; AdamW (learning rate 0.001, weight decay 0.001); StepLR (step 5, gamma 0.5); GRU hidden size 128; dropout 0.3.

The recorded command is: python -m src.compare_single_frame --device mps --epochs 15 --batch-size 4 --seed 42 --architecture both --output results/metrics/professor_comparison_v2. This describes the recorded run, not a guarantee that every hardware/software environment reproduces identical floating-point results.

5. Experimental Design and Evaluation

The study uses the same eligible 119-kick evaluation population and grouped folds for all methods and horizons. For each configuration, the final summary pools held-out predictions across five folds, giving 119 out-of-fold test predictions per horizon/model combination. Each kick contributes one held-out prediction per configuration. The neural configurations use pretrained weights, and the requested run was marked complete in the verified result file.

Accuracy is the proportion of correctly classified kicks. Precision measures the fraction of predictions for a class that are correct; recall measures the fraction of true examples of that class that are detected; F1 is their harmonic mean. Macro-F1 averages class F1 values without weighting by class frequency. Confusion matrices use rows for true class and columns for predicted class in the order left, right. These metrics are reported because accuracy alone obscures class-specific failures.

6. Quantitative Results

6.1 Pooled out-of-fold performance

Horizon	Model	Accuracy	Macro-F1
1.0 s	Majority	55.46%	35.68%
1.0 s	Pose logistic	36.97%	36.69%
1.0 s	MobileNetV3 + GRU	45.38%	44.61%
1.0 s	Single-frame MobileNetV3	47.90%	46.22%
0.5 s	Majority	55.46%	35.68%
0.5 s	Pose logistic	57.14%	57.13%
0.5 s	MobileNetV3 + GRU	50.42%	50.29%
0.5 s	Single-frame MobileNetV3	53.78%	53.31%
0.2 s	Majority	55.46%	35.68%
0.2 s	Pose logistic	63.87%	63.36%
0.2 s	MobileNetV3 + GRU	53.78%	51.46%
0.2 s	Single-frame MobileNetV3	51.26%	49.99%

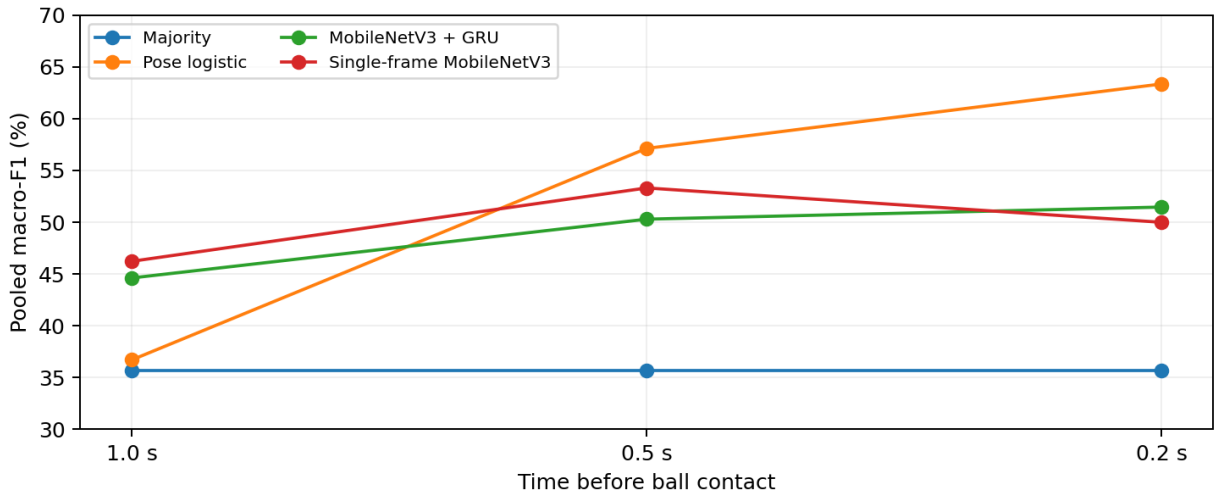


Figure 1. Pooled macro-F1 across the three pre-contact prediction horizons (five grouped folds; 119 held-out predictions per configuration).

6.2 Instructor-requested paired comparison

Horizon	GRU − single-frame accuracy	GRU − single-frame macro-F1
1.0 s	−2.52 percentage points	−1.61 percentage points
0.5 s	−3.36 percentage points	−3.01 percentage points
0.2 s	+2.52 percentage points	+1.48 percentage points

The GRU trails the single-frame model at 1.0 and 0.5 seconds, then leads narrowly at 0.2 seconds. This does not establish a consistent temporal advantage or statistical significance. Across the full set of models, pose logistic regression at 0.2 seconds is strongest on both accuracy and macro-F1.

6.3 Error analysis from confusion matrices

Model at 0.2 s	True left → left/right	True right → left/right
Pose logistic	45 / 21	22 / 31
GRU	45 / 21	34 / 19
Single-frame	40 / 26	32 / 21
Majority	66 / 0	53 / 0

The majority predictor correctly classifies all left kicks but none of the right kicks. At 0.2 seconds the GRU correctly detects 45/66 left kicks (68.18% recall) but only 19/53 right kicks (35.85% recall), suggesting an asymmetric failure pattern. The pose-logistic baseline is more balanced, detecting 45/66 left kicks and 31/53 right kicks. These counts are direct evidence of classification errors; without examining individual clips, they do not identify whether camera angle, occlusion, motion blur or player identity caused specific failures.

6.4 Timing

Horizon	GRU (s/kick)	Single frame (s/kick)
1.0 s	0.0545	0.0455
0.5 s	0.0536	0.0338
0.2 s	0.0530	0.0336

These measurements include decoding 16 frames and other measured pipeline work; they are not isolated neural-network latency. Timing comparisons should be interpreted with this measurement boundary in mind.

7. Discussion

The findings challenge the assumption that a more expressive video architecture must outperform a simpler model. The strongest overall method is the pose-logistic baseline at 0.2 seconds, reaching 63.87% accuracy and 63.36% macro-F1. One plausible explanation is that the pose representation concentrates on relevant kicker geometry, whereas a frozen generic image encoder may not emphasize subtle penalty-specific cues. This explanation is a hypothesis, not a demonstrated causal mechanism.

The neural training logs also show instances where training loss decreases while validation macro-F1 fluctuates or deteriorates. Such behavior is compatible with unstable generalization on a small dataset; it is not proof of a specific overfitting mechanism without additional diagnostics. The lack of a consistently superior GRU is a meaningful experimental result and meets the rubric's emphasis on controlled comparison and evidence-based interpretation.

8. Limitations and Future Work

The central limitation is dataset size, especially the effectively absent center class. Further concerns include imperfect grouping metadata, possible source overlap, annotation noise around ball-contact timing, domain shifts between matches or cameras, and uncertainty around generalization to unseen players. The experiment does not report confidence intervals or formal paired significance testing, and no claim of significance is warranted. Differences in classifier-head capacity also limit the strength of a temporal-only interpretation.

Next steps include acquiring a substantially larger and independently verified three-class dataset; auditing clip provenance and duplicate footage; checking player/match group IDs; documenting the exact preprocessing transforms and hyperparameters; comparing matched-capacity temporal and non-temporal heads; and performing systematic error inspection with uncertainty estimates. These are proposed improvements, not experiments already completed.

9. Demonstration and Reproducibility

A 60.00-second MP4 demonstration was generated locally at results/demo/soccer_penalty_final_demo.mp4. The video presents a kick labeled as held-out test kick 2-2, reports a left prediction against a right outcome, and transparently marks that prediction incorrect. The demo-generation script was corrected in commit 54dfd26 to display the verified 0.2-second GRU accuracy of 53.78%. The duration and ZIP integrity were checked from terminal output; the visual playback, audio, and held-out membership should be reviewed against the saved folds before submission.

The code and lightweight results are published in the project repository. The results folder includes summary.json, comparison.csv, folds.json, run_config.json, macro_f1_comparison.png, FINAL_RESULTS.md and SHA256SUMS.txt. Large checkpoints, raw videos and full fold-level predictions remain local. The recorded automated test run passed 10 tests. The source commit used for the comparison is 47a9824; the later results-publication commit is e13c942. A reproducer must obtain the underlying data independently and follow the README and environment instructions.

10. Conclusion

This project demonstrates a complete, quantitatively evaluated deep-learning workflow for pre-contact penalty-direction prediction. On the available 119-kick binary dataset, a pose-based logistic baseline outperformed both neural configurations at the closest pre-contact horizon. MobileNetV3-Small + GRU improved slightly over the single-frame variant at 0.2 seconds, but performed worse at 1.0 and 0.5 seconds. The appropriate conclusion is not that temporal modeling never helps; rather, these results do not show a consistent benefit under the evaluated data and model constraints. The project highlights the importance of baselines, group-aware evaluation, class-specific metrics and candid reporting of limitations.

References

[1] A. Howard et al., “Searching for MobileNetV3,” Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 1314–1324, 2019. doi: 10.1109/ICCV.2019.00140.

[2] K. Cho et al., “Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation,” Proceedings of EMNLP, pp. 1724–1734, 2014. doi: 10.3115/v1/D14-1179.

[3] A.-A. Opakunle and Z. Butz, Soccer-Penalty-Direction-Prediction, project source and experimental artifacts, GitHub repository, source commit 47a9824, results commit e13c942, and corrected demo script commit 54dfd26, 2026.

[4] F. Li and N. D. Pifer, “Women’s Soccer Penalty Kick Data Captured by Computer Vision Models,” Mendeley Data, version 2, 2026, doi:10.17632/brx9bsxnpx.2 (CC BY 4.0).

[5] S. Giancola et al., “SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos,” CVPR Workshops, 2018. Background only; SoccerNet footage is not asserted to be part of the evaluated 119 kicks.